

SELF FORMULATION PROTOCOL: PART III - IMPACT OF LIPOLYSIS ON PROTOTYPE FORMULATIONS

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1 - PURPOSE

The aim of this study was to develop a sound formulation protocol for Self Emulsifying Lipid Formulation (SELF) with the latest characterization tools proposed by the LFCS Consortium [1,2]. In this third and last part, we evaluated the ability of prototype formulations to maintain model drugs in solution after dispersion and *in vitro* lipolysis in fasted state.

2 - METHODS

The therapeutic dose of each model drug (200 mg of Ibuprofen, 25 mg of Cinnarizine, and 10 mg of Piroxicam) is loaded in 1 mL of prototype formulations. Table 1 summarizes lipid-based formulation types selected and the percentage of saturation of the drug in each formulation.

• Preparation of media and enzyme suspension

The lipolysis buffer is prepared by adding Tris (0.474 g/L), CaCl₂ (0.208 g/L) and NaCl (8.810 g/L) in Milli-Q water. The pH is adjusted at 6.5 with either HCl 1M or NaOH 0.6 M. The lipolysis medium is prepared by adding L-α-phosphatidylcholine (0.576 g/L) and NaTDC (0.576 g/L) in the lipolysis buffer. The medium is stirred overnight to allow the complete solubilisation of lecithin.

The pancreatin solution is prepared by adding 1 g of pancreatin powder in 5 mL of lipolysis buffer. After 10 minutes of magnetic stirring the solution is centrifuged (Universal centrifuge 320R) at 2800g and 5°C for 10 min. The supernatant is sampled to be used in the lipolysis test.

• In vitro lipolysis test

The experimental setup consists in a pH stat apparatus (Metrohm® AG, Switzerland), comprising a Titrand 802 propeller stirrer/804 Ti Stand combination, a glass pH electrode (iUnitrode) and one 800 Dosino dosing units coupled to 5 mL autoburette. The apparatus is connected to a PC and operated using Tiamo 2.0 software. The equipment is thermostated at 37°C and filled with lipolysis medium, to mimic fasted state in the small intestine. During the digestion of lipid-based formulations by pancreatic lipases, pH is maintained constant by the addition of either NaOH 0.1N for formulations containing long chain fatty acids (Gelucire® 48/16 for example) or NaOH 0.6N for formulations containing medium chain fatty acids (Labrasol® ALF for example).

Before the lipolysis test, the specific activity of pancreatic lipases is checked on tributyrine (model substrate). The substrate must be in excess to have a reaction rate directly proportional to the enzyme concentration. Five hundred microlitre of tributyrine and 1.5 µL of pancreatin solution are introduced in 28 mL of lipolysis medium (without lecithin). The specific activity of pancreatic lipases should be higher than 900 U/mg.

One gram of lipid-based formulations in hard gelatine capsules (Licaps, Capsugel), is added to 36 mL of lipolysis medium at 37°C. After 10 minutes (dispersion / emulsification time), 4 mL of pancreatin solution is added. The duration of the test is 1 hour. Aliquots of 1 mL are sampled at various time points to determine the solubility of the drug in the micellar phase: t=-5 min (during the dispersion time), t=0 min (before the addition of the pancreatic solution), t=5 min, t=15 min, t=30 min and t=60 min.

Five microlitres of the inhibitor solution (1M: 100 mg of 4-bromophenylboronic acid in 0.5 mL of methanol) is added immediately to each sample. They are then centrifuged at 21000g and 37°C for 30 minutes. The supernatant is sampled and diluted with an appropriate solvent (a solution of valerophenone in acetonitrile (0.35 mg/mL) for Ibuprofen, and acetonitrile for Cinnarizine and Piroxicam) for further HPLC analysis.

Placebo formulations are also tested in order to determine the solubility of the drug in the lipolysis medium containing the digested lipid formulation. In this case only one sampling is performed at t=60 min. The micellar phase is obtained at the end of lipolysis experiment after centrifugation at 21000g and 37°C for 30 minutes of aliquots of the pH-stat medium. The supernatant is sampled and drug is added in excess. Mixtures are let at 37°C under agitation for 4 hours. Aliquots are then centrifuged as described above and diluted with the appropriate solvent for HPLC analysis.

3 - RESULTS AND DISCUSSION

• Piroxicam formulations

The ease of emulsification of Piroxicam formulations is clearly related to the physical state of excipients and their ability to form mesophases. Labrasol® ALF is liquid and self-emulsifies in a few seconds, allowing the complete dissolution of Piroxicam in the aqueous phase in less than 5 minutes. This is the only excipient able to dissolve and maintain the 10 mg of Piroxicam within the aqueous phases during the dispersion phase. Tween® 80 is slower as it is a viscous liquid that forms viscous mesophases when in contact with water. Gelucire® 44/14 and Cremophor® RH40 are even slower as they are semi-solid excipients needing more time to hydrate and form viscous mesophases that erode slowly. Finally, the new Gelucire® 48/16, a solid surfactant design for solid oral dosage forms, is even slower due to its higher melting point.

These formulations composed of water-soluble surfactants able to dissolve the dose of Piroxicam showed comparable behaviour after digestion even though saturation levels were different. Formulations containing medium chain fatty acids (Labrasol® ALF) are more digested than formulations with long chain, but can in this case keep the drug in solution more efficiently. The turbid dispersion obtained with Labrasol® (67% of saturation) is more efficient in dissolving and maintaining Piroxicam in the aqueous phase than other surfactants leading to micellar solution (for example Cremophor® RH40 (35% of saturation), Figure 1). It should be noted that even if Gelucire® 48/16 is slower to erode because of its higher melting point, it leads to the higher and more constant concentration of Piroxicam in the aqueous phase during the 1-hour digestion.

• Cinnarizine formulations

Solution of Cinnarizine with Labrasol® ALF led to incomplete release of the drug in the micellar phase and a rapid precipitation after the first minutes of digestion (Figure 2). Adding a hydrophilic cosolvent such as Transcutol® HP to increase the solvent power of the formulation gave the same result. This is due to the low solubilising capacity of the digested micellar phase containing Labrasol®. Table 1 presents the solubility of the drug within the digested micellar phases and the corresponding maximum supersaturation ratio (SR^M) calculated with Equation 1. When this ratio is higher than 3, significant precipitation of the drug is expected as explained in [2].

$$\text{Equation 1} \quad SR^M = \frac{\text{Maximum drug solubility in AP}}{\text{Drug solubility in AP digested}}$$

SR^M: maximum supersaturation ratio
AP: aqueous phase

Cinnarizine is known to have a good solubility in mixed micelles containing oleic acid [3]. Hence 15% of Maisine™ 35-1 was added to a Labrasol®-based formulation even if this lipid excipient did not show good solvent capacity during the initial solubility screening. After dispersion, the concentration of cinnarizine dissolved in the aqueous phase is equivalent to the previous formulation, but after 5 minutes of digestion, the release of oleic acid by Maisine™ 35-1 led to a dramatic increase of the solubility of Cinnarizine within the aqueous phase (Figure 2). However due to the low stability of the mixed micelles this concentration steadily decreases during the following hour. This case study shows the importance of measuring the solubility of the drug within the digested micellar phase and calculating the SR^M to select the good excipient and anticipate the impact of supersaturation.

• Ibuprofen formulations

All formulations led to low supersaturation SR^M ≤ 3 and no significant precipitation was observed (Figure 3). This confirms the utility of SR^M as an indicator of the ability of the formulation to keep the drug in solution after *in vitro* digestion.

Table 1. Selected prototype formulations for Ibuprofen, Cinnarizine, and Piroxicam – percentage of saturation, solubility in digested aqueous phase and supersaturation ratio.

Drug	Prototype formulations	LFCS Type	%Sat	Solubility in digested aqueous phase (mg/mL)	SR ^M
Ibuprofen	94% LAS / 6% TRA	IIIB	61	2.20	2.5
	94% 4816 / 6% TRA	IV	86	1.75	3.2
	94% T80 / 6% TRA	IV	75	2.69	2.1
	100% 4816	IV	67	2.01	2.8
	100% 4414	IIIB	67	1.90	2.97
Cinnarizine	90% LAS/10% C90	IIIB	72	0.02	43.4
	75% LAS/10% C90/15% TRA	IIIB	105	0.02	43.4
	75% LAS/10% C90/15% MAI	IIIB	63	1.08	0.6
Piroxicam	100% LAS	IIIB	67	ND	ND
	100% T80	IV	54	ND	ND
	100% RH40	IV	35	ND	ND
	100% 4816	IV	50	ND	ND
	100% 4414	IIIB	50	ND	ND

with: LAS=Labrasol® ALF, TRA=Transcutol® HP, 4816= Gelucire® 48/16, T80=Tween® 80, 4414=Gelucire® 44/14, C90=Capryol™ 90, RH40=Cremophor® RH40, %Sat=percentage of saturation, SR^M=maximum supersaturation ratio, ND=not determined.

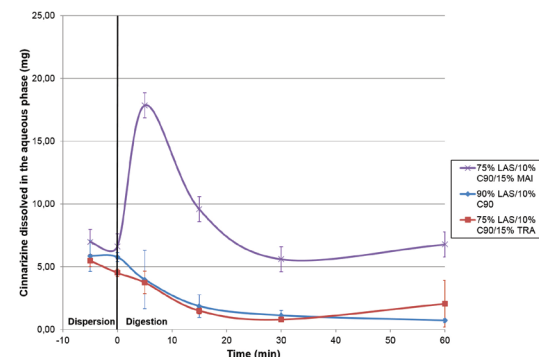


Figure 2. Evolution of the weight of Cinnarizine dissolved in the aqueous phase during in vitro dispersion and digestion.

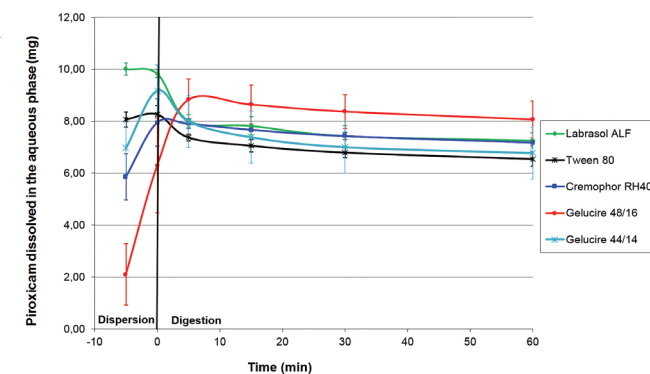


Figure 1. Evolution of the weight of Piroxicam dissolved in the aqueous phase during in vitro dispersion and digestion.

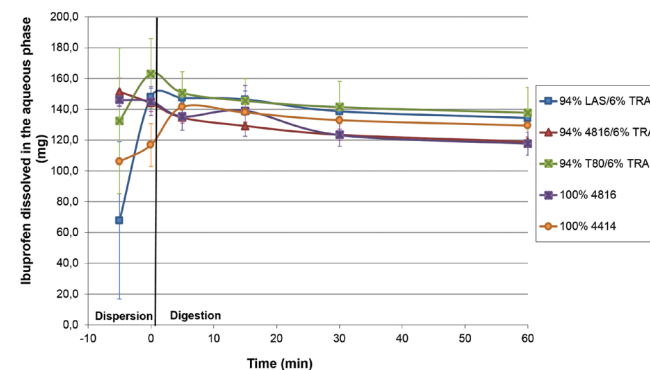


Figure 3. Evolution of the weight of Ibuprofen dissolved in the aqueous phase during in vitro dispersion and digestion.

4 - CONCLUSIONS

During *in vitro* lipolysis test simulating small intestine fasted state, the percentage of saturation and the extent of digestion (volume of fatty acid titrated) are not always predictive parameters to compare formulations. The dosage of the drug in the micellar phase is mandatory to assay the ability of the formulation to maintain drug in solution. This study also demonstrates that turbid formulations consisting of digested excipients can be more efficient than supposed non-digestible micelle-forming formulations.

5 - REFERENCES

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